

CS-499: 7-1 Professional Self-Assessment

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Throughout these last few years in the Bachelor of Science in Computer Science program at Southern New Hampshire University, I have learned and implemented both technical skills and a professional mindset that will be necessary to begin my career as a software developer. As I went into a new course, it introduced new concepts and technologies; this capstone, and seeing my ePortfolio come together, is what allowed me to reflect on those individual experiences, and see everything fit together, and show me the progress I have made during the program. When I take a look at my previous coursework and compare that to the work I am showing in this portfolio, it seems almost surreal how big of a jump that is, and I can see how my understanding has matured beyond writing functional code, into actually designing applications which are maintainable, secure, scalable, and built using software engineering best practices.

I think being able to communicate and collaborate with different audiences is one of the most valuable skills I have developed throughout the program, even if many of the projects completed in the course were individual assessments with no group work being done, they simulated real world software development environments where clear communication is needed, and that was shown through reports and narratives which I wrote alongside every modules technical submission. Throughout these courses, I created technical reports, software design documents, presentations, and recorded code reviews, which had me explaining technical concepts in a clear format where my goal was to explain technical concepts clearly and demonstrate my understanding of the decisions I made, and why they were right, at least to my understanding. This capstone strengthened those communication skills by requiring professional narratives, alongside every enhancement submission, which was used to justify design decisions and how specific enhancements aligned with the Computer Science program outcomes. I have concluded that learning to communicate technical information is just as important as writing the code itself, especially when working with project managers, clients, stakeholders, and other software developers.

Courses such as CS465 were a start to full-stack web development, which was made using the MEAN stack and also gave me an idea of architectural concepts such as MVC, RESTful APIs, authentication, and modular application design. Earlier software courses either strengthened or introduced me to object-oriented programming, testing, debugging, documentation, and maintainable code organization; then in this capstone, I had the chance to revisit these concepts while being tasked with enhancing an existing application, where I had to improve overall architecture instead of simply adding new functionality. With this experience, I learnt that professional software development is a consistent process where applications are constantly improved and refined over time. When I glanced at the courses for this program, seeing capstone, I assumed I would simply make another application, but never did I think I would reopen a project I once considered completed, to try and make it better than it once was.

Earlier course work gave me an introduction to algorithms and data structures, where I did touch on concepts of searching, sorting, recursion, algorithm efficiency, and data organization, and now during this capstone I was able to apply those same principles practically for a full-stack application, where I got to implement backend searching, dynamic sorting, and pagination using MongoDB queries, as opposed to using algorithms as an isolated exercise, I got to use it directly and see how it would influence applications performance, and user experience. This helped to grow my understanding of algorithms and data structures, and showed me the importance of selecting appropriate solutions while balancing usability.

Earlier course work gave me a base understanding of databases, queries, and data management, while courses later on gave me experience connecting applications to databases, which is something that I had to work on during the capstone as well. In the capstone, I was able to build on that foundation I initially had by improving the MongoDB database used by the Travlr application. This was done by implementing a stronger schema validation, indexing, data normalization, and query optimization. By doing this, along with my prior knowledge of databases, I now know that database development isn't just storing and retrieving information, but it's also how to make sure that data remains safe, consistent, and reliable for when the application grows to more than just a few trips.

Another concept which was reinforced during the capstone was developing a security mindset. We had a course prior that touched on software concepts, secure software development, and vulnerability assessment, but similar to other classes, what we learnt was used in isolated assessments, rather than a full-stack application. So the concepts I learnt were reinforced during the capstone through the addition of JSON Web Token (JWT) authentication, protected administrative routes, better backend validation, as well as secure handling of user input. Doing this work, I noticed how security isn't a feature that's added near the end of development, like 'Oh my application is done, let me make it secure now,' but something that is done throughout and should influence every layer of an application, from the database to the interface.

Beyond the artifact I chose to include for this portfolio and capstone, I was exposed to a broad range of technologies, tasks, and problem-solving methods in this Computer Science program. A few examples are: the software testing we did in CS320, where I gained experience with automated unit testing, and how, in CS305, I strengthened my understanding of secure software development through concepts such as vulnerability analysis. In other courses, I was able to explore more specialised areas of computer science like creating interactive three-dimensional graphics in CS330 and working with artificial intelligence concepts and reinforcement learning in CS370. Although these areas are not directly related to the artifact presented in my ePortfolio, they broadened my understanding of computer science and showed me how many different areas of the field can contribute to solving software problems. Outside of just the academic coursework I have completed, I have also been studying for the AWS Certified Cloud Practitioner certification, as well as attempting to complete other certificates, to expand the understanding I already have, especially in areas like cloud computing and modern application infrastructure.

The artifacts included within this ePortfolio come together to showcase the progression of both my technical abilities as well as my professional growth throughout not only this class, but my Computer Science program as a whole. Instead of choosing unrelated assignments from different courses, my portfolio shows the progress and evolution of one single full-stack application, across three enhancement milestones. The original artifact shows the foundation on which each enhancement was built, and eventually stacked on top to have a fully completed piece of software. Initially, the enhancements started with software engineering by strengthening authentication, validation, reusable controller logic, and the application architecture as a whole, where then we moved onto the second enhancement, where my understanding of algorithms and data structures was shown, where I implemented backend searching, sorting, and pagination, which I also think would be my favourite enhancement out of the three. The final enhancement wasn't as front-end as the others, so the changes don't look as big, but it focuses on database development, where I went in with the plan of improving schema validation, indexing, and query optimization, which was done to create a more reliable application.

Together, these enhancements taught me that professional software development isn't only writing lines of code that function correctly. Each milestone made me analyze an existing application and try to find opportunities for improvement, on a piece of software I once considered to be perfect and completed. I had to evaluate trade-offs and implement meaningful enhancements that went far beyond making the application look visually appealing. The progression shown from enhancement to enhancement in this portfolio shows both the technical knowledge that I have gained during the Computer Science program, but it shows how I have learnt how to incorporate that knowledge within a realistic software project.

Being closer than ever before to completing this degree and developing an ePortfolio has me confident about entering the real world for a professional career. Although the software industry is competitive and continues to change rapidly, I am confident that this program has provided me with the foundation I need to begin my career and continue developing my skills. This program has also strengthened my problem-solving abilities, helped me understand that the computer science field is full of continuous learning, and provided me with experience applying industry best practices. I believe that this portfolio represents not only the technical skills I have learnt from this program, but also the commitment and dedication I have to produce reliable, maintainable, and secure software solutions while continuing to grow as a computer science professional.